Department of ICT

Session: 2023-24

Medique

Smart Clinic Appointment & Queue Management System

Capstone Project Proposal

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1 Abstract

Long queues and manual appointment systems cause significant waiting time and patient dissatisfaction in clinics. **Medique** is a web-based smart clinic appointment and queue management system that digitizes the complete patient journey: searching for doctors, booking appointments, receiving a token number, and tracking live queue position with an estimated waiting time.

The system serves three roles—Patient, Doctor, and Admin—through separate dashboards. It is designed using object-oriented principles with ten core classes: User, Patient, Doctor, Admin, Appointment, Queue, DoctorSchedule, Notification, Department, and Clinic. The prototype is built with HTML5, CSS3, and JavaScript and demonstrates the full booking-to-queue flow using realistic mock data.

The expected outcome is a working front-end prototype, complete UML documentation, and a project plan that shows how Medique reduces perceived waiting time and improves clinic workflow.

2 Introduction

Clinics today still rely on walk-in appointments, paper tokens, and manual register books. Patients have no way to know how long they must wait, and clinic staff spend excessive time managing queues instead of caring for patients. Medique solves this by moving the appointment and queue process online.

Patients can search doctors, view availability, book a time slot, and receive a unique token (e.g., `A-27`). A live queue tracker shows the current token, patients ahead, and estimated wait. Doctors get a queue control panel (Call Next / Mark Served / Skip), and administrators get management and reporting tools.

This proposal covers the problem, objectives, scope, requirements, design (UML), schedule, and the prototype plan.

3 Problem Statement

Current clinic workflows suffer from:

- **Unpredictable waiting times** — patients wait for hours with no information.
- **Manual token systems** — no live tracking, frequent confusion and crowd buildup.
- **Scheduling errors** — double bookings and missed appointments.
- **No visibility** — patients cannot see doctor availability or their queue position.
- **Administrative burden** — staff manually manage appointments and records.
- **No data** — clinics cannot analyze performance or improve services.

These problems reduce patient satisfaction, waste doctor time, and limit the clinic's ability to grow.

4 Objectives

General Objective

Design and develop a professional web prototype of Medique that demonstrates smart appointment and live queue management.

Specific Objectives

1. Design a system with at least 10 OO classes and their relationships.
2. Develop a responsive front-end prototype (HTML5/CSS3/JavaScript) covering patient, doctor, and admin flows.
3. Implement a token-based queue tracker with live position and estimated wait.
4. Produce complete UML and DFD documentation.
5. Demonstrate the end-to-end flow: select doctor → select date → select time → book → receive token → track queue.

5 Scope

In Scope

- Patient: registration, login, search doctors, book/cancel appointment, view token, track queue, view history, notifications.
- Doctor: login, set schedule, view today's appointments, control queue (call/skip/serve), view history.
- Admin: manage patients, doctors, departments, schedules, appointments, queues; publish announcements; view statistics.
- Smart queue: current token, patient token, patients ahead, estimated wait, status.

Out of Scope (Prototype Phase)

- Backend server and real database (prototype uses mock data).
- Real payment, SMS/email integration, telemedicine, prescriptions.
- Multi-clinic support and native mobile apps.

6 Target Users

- **Patients (primary)** — book appointments, track queues, reduce waiting.
- **Doctors (secondary)** — manage schedules, control patient flow daily.
- **Admins / Clinic staff (secondary)** — oversee operations, manage users, generate reports.

7 Functional Requirements

ID	Requirement	Description
FR-01	Patient Registration	Register with name, email, password, phone.
FR-02	Login / Logout	Authenticate all users with role-based access.
FR-03	Doctor Search	Search and filter doctors by specialization and availability.
FR-04	Doctor Profile	View specialization, experience, fee, schedule.
FR-05	Book Appointment	Select doctor → date → time slot → confirm.
FR-06	Token Generation	Generate unique appointment ID and token (e.g., A-27).
FR-07	Queue Tracking	Show current token, patient token, patients ahead, estimated wait, status.
FR-08	Cancel Appointment	Patient can cancel upcoming appointments.
FR-09	Appointment History	View past appointments with status (Upcoming, Waiting, Completed, Cancelled).
FR-10	Doctor Queue Control	Call Next, Mark Served, Skip patient.
FR-11	Doctor Schedule	Doctor sets available days and time slots.
FR-12	Admin Management	Manage patients, doctors, departments, schedules, appointments.
FR-13	Announcements	Admin publishes notices visible to patients.
FR-14	Reports	Admin views statistics and charts on clinic activity.

8 Non-Functional Requirements

Category	Requirement	Description
Performance	Response time	UI responds within 2 seconds.
Performance	Queue refresh	Queue data updates within 5 seconds.
Usability	Intuitive UI	Clear navigation, consistent components.
Usability	Responsive	Works on desktop, tablet, and mobile.

Category	Requirement	Description
Accessibility	WCAG 2.1 AA	Sufficient contrast, readable fonts.
Security	AuthN / AuthZ	Role-based access for all operations.
Reliability	Availability	99.5% uptime target (future full system).
Maintainability	Modular code	Well-commented, component-based structure.

9 Proposed Solution

Medique is a three-portal web application:

1. **Patient Portal** — search doctors, book appointments, receive token, track live queue, view history.
2. **Doctor Portal** — view today's queue, call next, mark served, skip, manage schedule.
3. **Admin Dashboard** — manage users and departments, monitor queues, publish notices, view reports.

The core innovation is the **smart token queue**. Each booking produces a token (e.g., `A-27`). Estimated wait is computed as:

$$EstimatedWait = (PatientsAhead) \times (AverageConsultationTime)$$

This gives patients realistic expectations and helps clinics plan better.

10 Software Process Model

Iterative and Incremental Model is chosen because:

- Requirements can be refined after each increment.
- Working prototype is available early for feedback.
- Risks are identified and resolved earlier.
- Fits a semester-based capstone timeline well.

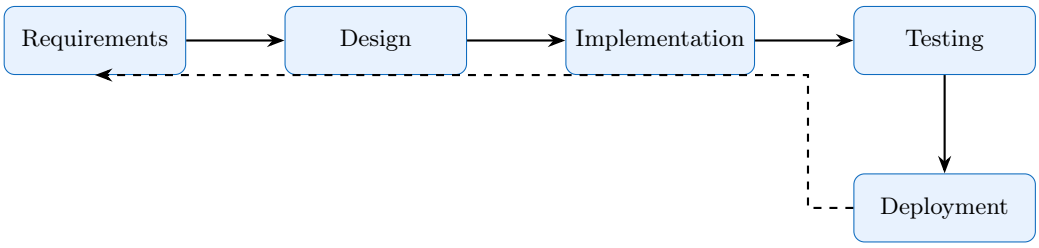


Figure 1: Iterative and Incremental Process Model

11 Project Schedule

Week	Activity	Iteration
1	Requirements gathering, literature review	Iteration 1
2	System design, UML diagrams	Iteration 1
3	Proposal completion, UI mock-ups	Iteration 1
4	Landing page and authentication UI	Iteration 2
5	Doctor search and profile pages	Iteration 2
6	Appointment booking flow	Iteration 2
7	Queue tracking with live updates	Iteration 3
8	Patient and Doctor dashboards	Iteration 3
9	Admin dashboard and reports	Iteration 3
10	UI/UX refinement and testing	Iteration 4
11	Documentation and bug fixes	Iteration 4
12	Final presentation preparation	Iteration 4

12 Milestones

M#	Milestone	Output	Week
M1	Requirements & Design complete	UML docs, Proposal	3
M2	Core UI prototype	Landing, Auth, Search	6
M3	Booking & Queue functional	Booking, Token, Queue	9
M4	Complete prototype	All screens, interactions	11
M5	Final submission	Docs + presentation	12

13 Deliverables

1. Project proposal document (this PDF).
2. Working front-end prototype (HTML/CSS/JS, GitHub).

3. UML and DFD documentation (Context DFD, Level-1 DFD, Use Case, Activity, ER, Class, Object diagrams).

4. Source code with comments and README.

5. Final presentation slides and live demo.

14 Gantt Chart

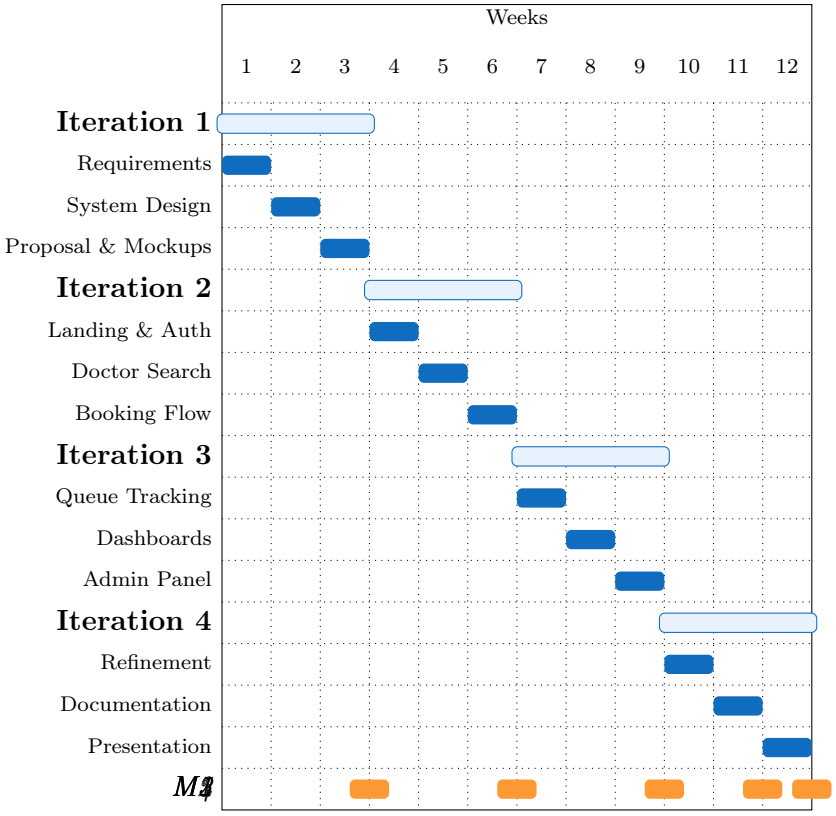


Figure 2: Medique Gantt Chart (12 Weeks)

15 System Architecture

Medique follows a three-tier architecture.

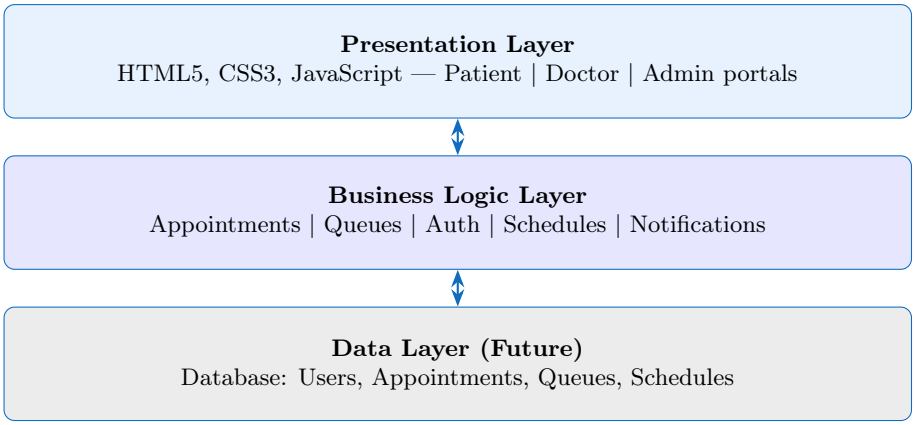


Figure 3: Three-Tier Architecture

16 Context Level DFD (Level 0)

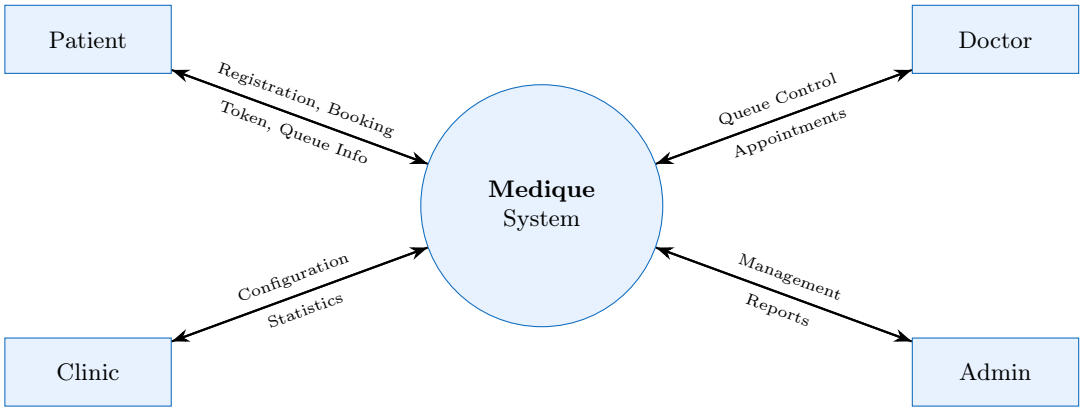


Figure 4: Context Level DFD

17 Level-1 DFD

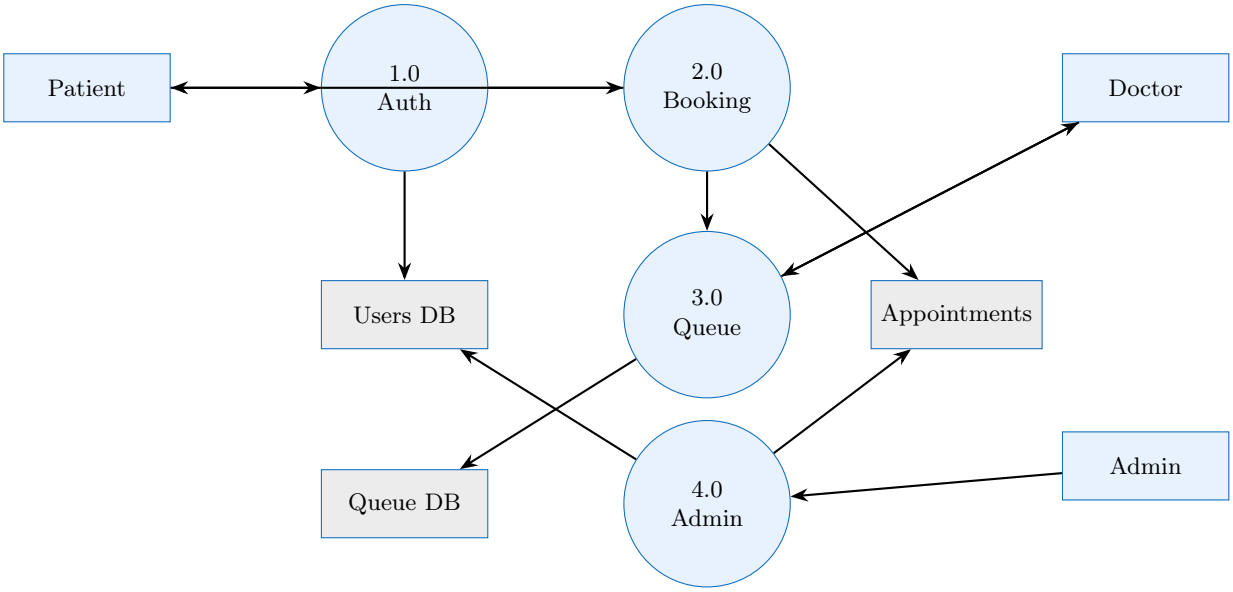


Figure 5: Level-1 DFD

18 Use Case Diagram

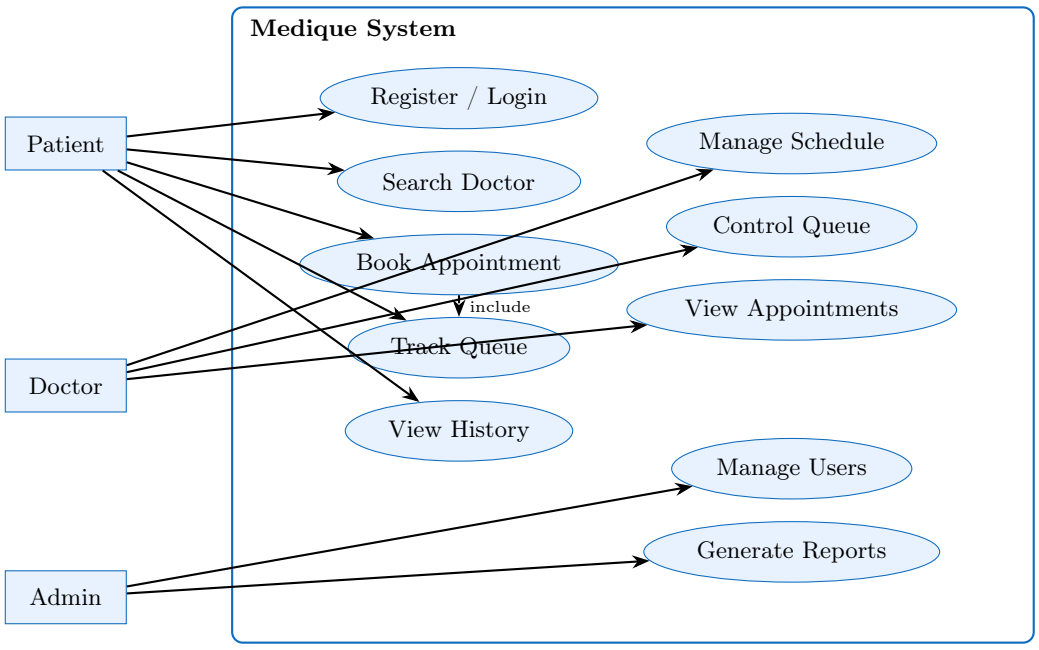


Figure 6: Use Case Diagram

19 Activity Diagram

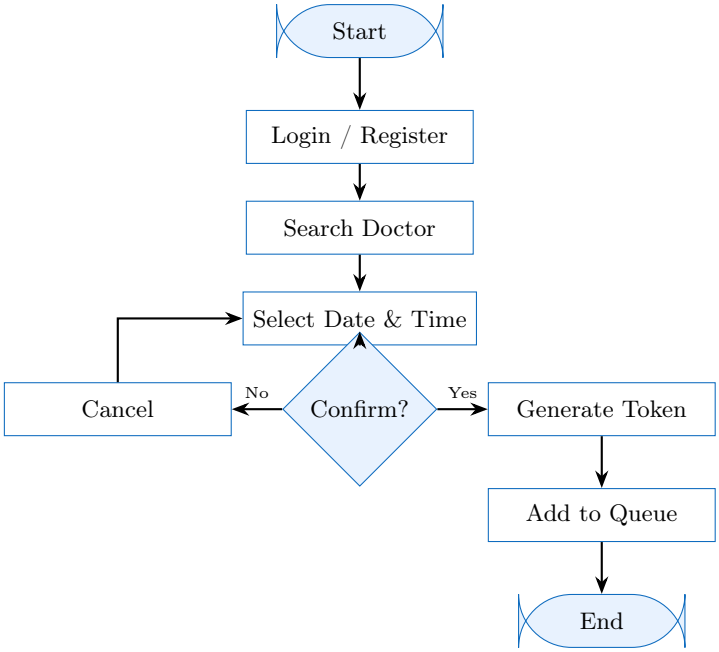


Figure 7: Activity Diagram — Appointment Booking

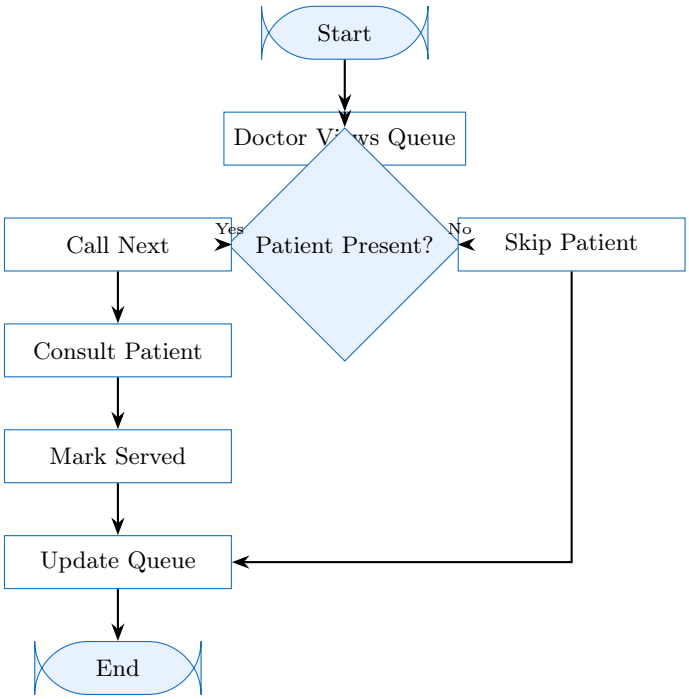


Figure 8: Activity Diagram — Queue Management

20 Entity Relationship Diagram

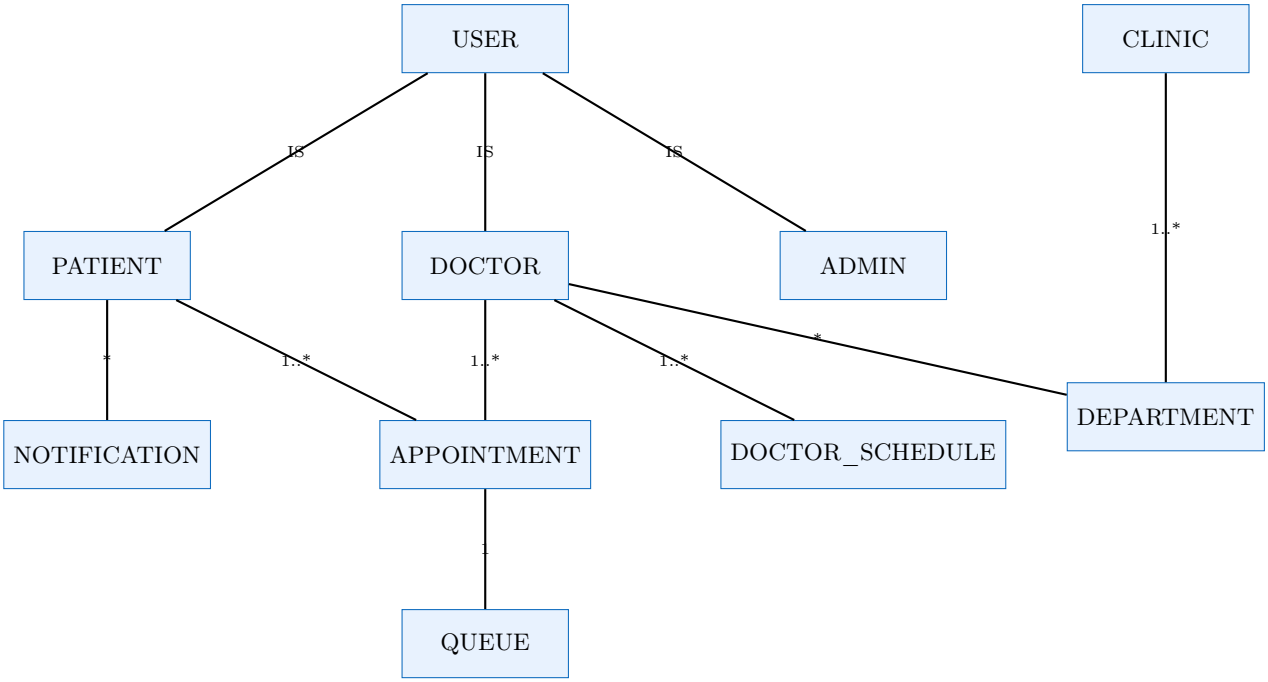


Figure 9: Entity Relationship Diagram

Entity	**Key Attributes**
USER	user_id (PK), email, password, role, name, phone, created_at
PATIENT	patient_id (PK), user_id (FK), dob, address
DOCTOR	doctor_id (PK), user_id (FK), specialization, experience, fee, dept_id (FK)
ADMIN	admin_id (PK), user_id (FK), permissions
CLINIC	clinic_id (PK), name, address, phone
DEPARTMENT	dept_id (PK), name, clinic_id (FK)
APPOINTMENT	appointment_id (PK), patient_id (FK), doctor_id (FK), date, time, token_number, status, created_at
QUEUE	queue_id (PK), appointment_id (FK), token_number, queue_status, current_position, estimated_waiting_time
DOCTOR_SCHEDULE	schedule_id (PK), doctor_id (FK), day_of_week, start_time, end_time
NOTIFICATION	notif_id (PK), patient_id (FK), message, type, is_read, created_at

21 Class Diagram

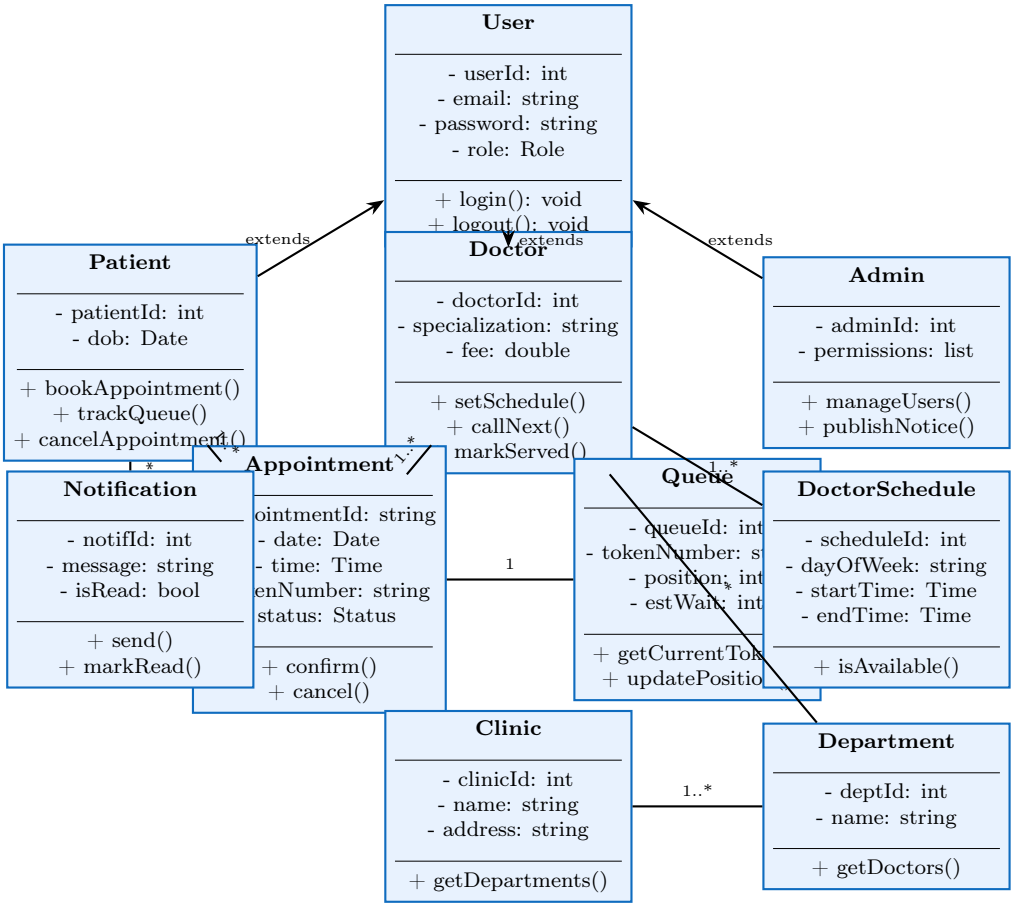


Figure 10: Class Diagram

22 Object Diagram

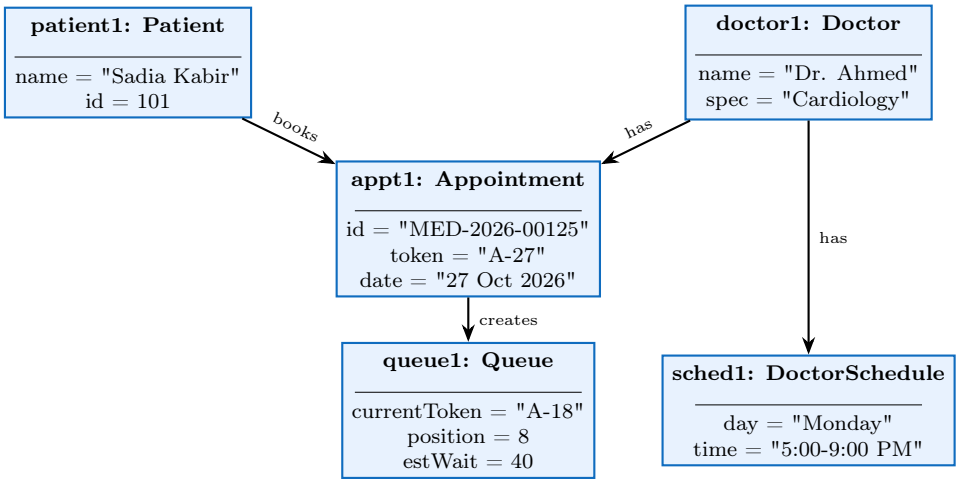


Figure 11: Object Diagram — Sample Appointment Instance

23 Prototype Screens

The front-end prototype (HTML5/CSS3/JavaScript) implements the following screens, each reflecting at least one of the ten classes:

#	Screen	Key Elements / Classes Reflected
1	Landing Page	Hero, Book Appointment, Find a Doctor, Features (*Clinic, Doctor*)
2	Login / Registration	Email/password, role select (*User, Patient, Doctor, Admin*)
3	Patient Dashboard	Upcoming appointment, token, queue position, wait (*Patient, Appointment, Queue*)
4	Find Doctor	Search, specialty filter, doctor cards (*Doctor, Department*)
5	Doctor Profile	Photo, specialty, fee, schedule, Book button (*Doctor, DoctorSchedule*)
6	Appointment Booking	Doctor → date → time → confirm → token (*Appointment, Patient, Doctor*)
7	Queue Tracking	Current token, your token, patients ahead, est. wait, status (*Queue, Appointment*)
8	Appointment History	Date, doctor, department, token, status (*Appointment, Doctor*)
9	Doctor Dashboard	Today's queue, current token, Call Next / Mark Served / Skip (*Doctor, Queue*)
10	Admin Dashboard	Stats cards (patients, doctors, appointments), chart, notices (*Admin, Department, Clinic*)

Demonstrated Workflow (5 classes)

The prototype demonstrates the end-to-end flow using five classes from the Class Diagram:

Patient selects **Doctor** → selects date/time via **DoctorSchedule** → **Appointment** is created with token → **Queue** position and wait time are displayed.

24 Technology Stack

Layer	Technology	Purpose
Structure	HTML5	Semantic markup
Styling	CSS3	Responsive design, Flexbox/Grid
Interactivity	JavaScript (ES6)	UI logic, queue simulation
Icons	Font Awesome 6	Healthcare UI icons
Fonts	Google Fonts (Inter)	Readable typography

Layer	Technology	Purpose
Version Control	Git / GitHub	Code hosting and submission
Documentation	LaTeX + TikZ	Proposal and diagrams

Planned for full version: Node.js/Express or Django (backend), PostgreSQL (database), JWT (auth), Socket.io (real-time queue).

25 Future Enhancements

- SMS and email appointment reminders.
- Real-time queue updates via WebSockets.
- Online payment for consultation fees.
- Telemedicine / video consultation.
- Digital prescription management.
- AI-based doctor recommendation.
- Emergency appointment prioritization.
- Native mobile application.
- Multi-clinic support.

26 Conclusion

Medique addresses a real clinic problem—unpredictable waiting time and manual queue handling—with a simple, modern web solution. This proposal defines the problem, objectives, scope, requirements, OO design (10 classes), complete UML and DFD documentation, and a 12-week iterative schedule with milestones and a Gantt chart.

The front-end prototype demonstrates the full patient journey using five core classes (Patient, Doctor, DoctorSchedule, Appointment, Queue) and provides dashboards for doctors and administrators. With a clear roadmap for future enhancements, Medique is a practical, scalable foundation for a real clinic management system.